Homework 8

Due: Friday, April 15

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

Problem 1

a. Andy, Michelle, and Katrina will engage in a tea fight to resolve dissent over Homework 7. They will take turns trying to splash tea on each other until only one person is left dry. They will go in the order (Andy, Michelle, Katrina), skipping anyone who has been splashed. They all know that Andy's chance of hitting his target is 0.3, Katrina's is 0.5, and Michelle never misses.

   Andy, who goes first, gets to choose which strategy to use on his first turn:

   - Try to splash Michelle
   - Try to splash Katrina
   - Deliberately miss both of them

   After this initial turn, all the other choices are random and no one can deliberately miss. (You can imagine that, after the first turn, if there are 3 people standing, the person whose turn it is flips a coin to determine whom to try to splash.)

   Show that Andy is most likely to win if he chooses to try to splash Michelle on the first turn.

b. (Extra Credit) If Andy and/or Katrina had a different probability of success, could it be possible for deliberately missing to be Andy's best strategy on his first turn? Justify your answer.

Problem 2

Consider the sample space Ω and events A, B, and C as follows.

$$\begin{aligned}\Omega &= \{a, b, c, d, e, f, g, h, i\}\\ A &= \{a, b, c\}\\ B &= \{c, d, e\}\\ C &= \{c, f, g\}\end{aligned}$$

Assume all outcomes are equally likely and that exactly one of them occurs.

a. Are A and B independent? B and C? A and C? Are events A, B, and C pairwise independent?
b. Are A, B, and C independent? Are events A, B, and C mutually independent?
c. Is A independent of $B \cup C$?

Problem 3

A CS22 student is working on a research paper but the local library has made it so that each person can only take out a single book. In front of the student is an aisle of n different books relevant to their paper. However, we do not know the value of n, only that n is a positive integer.

The student doesn't have enough time to go through the entire aisle, so they want to select a book **uniformly at random**. They use the following strategy as they walk down the aisle of books.

- Upon arriving at book 1, the student will choose it and put it in their bag.
- Upon arriving at book 2, the student will replace the current book in their bag with book 2 with probability $\frac{1}{2}$.

  $\vdots$

- Upon arriving at book i, the student will dump the current book and replace it with book i with probability $\frac{1}{i}$.

  $\vdots$

- Upon arriving at book n, with probability $\frac{1}{n}$ the student will dump the current book and replace it with book n.

Prove that by the time the student reaches the end of the aisle, the book they have will have been selected uniformly at random from all n books. In other words, a given book will be in the student's bag at the end of the aisle with probability $\frac{1}{n}$.

☕ Problem 4 (Mind Bender — *Extra Credit*)

The CS22 cafe has been converted to a casino for the evening. Rob is in charge, and offers guests the opportunity to play the following game:

- You pay Rob some amount of money, $\$x$, to start playing.
- Rob puts \$2 in the pot.
- Rob repeatedly flips a fair coin. Every time the coin comes up heads, he doubles the money in the pot. As soon as the coin lands on tails for the first time, the game is over.
- Once the game has ended, you receive all of the money in the pot.

Sounds more fun than roulette! Let's analyze this game with the language of probability that we've been developing.

a. Before reading any further, without doing any calculations, think about this question: how much money $\$x$ would you be willing to pay to play this game? (If you don't gamble, pretend you're James Bond.) Briefly explain why you decided on this value.

b. The description of this game hints at both a *sample space* and a *random variable.* Can you identify what they are? What are the outcomes that make up the sample space? What is the probability of each such outcome? Note that, even though almost all of our examples so far have concerned *finite* sample spaces, this one is infinite!

c. We defined the expected value of a random variable over finite sample spaces. But for countably infinite spaces, we can write down a similar definition: if we enumerate the outcomes $\omega_0, \omega_1, \omega_2, \ldots$, we can say for a random variable X that

$$\mathbb{E}(X) = \sum_{i=0}^{\infty} X(\omega_i) \cdot \mathbb{P}(\omega_i),$$

if this value exists. (It might not, or it might depend on the order in which we enumerate the ω_is. The question of when this sum makes sense is a topic for another course!)

Try to use this formula to compute the expected payoff from playing this game. What happens? Does this make you want to change your answer to part a?

This game is known as the *St. Petersburg paradox.* While it isn't a mathematical paradox—there's no contradiction—many people feel that the conclusion implied by the expected value calculation is different from how they would behave in reality. Philosophers and economists have proposed different resolutions to this paradox that try to explain the disconnect between math and reality. Read up on Wikipedia if you're curious!